

MITRE ATT&CK Mapping

The Linux STIG-style audit findings can also be mapped to several MITRE ATT&CK techniques that are commonly associated with Linux-based attacks and post-compromise activities. While this project focused on defensive auditing rather than active exploitation, the reviewed configurations and system artifacts relate directly to attacker persistence, privilege escalation, credential access, and system discovery behaviors.

MITRE Technique ID	Technique Name	Related Audit Activity
T1082	System Information Discovery	Collected OS version, hostname, kernel, and network configuration information
T1057	Process Discovery	Reviewed active running processes using ps aux
T1016	System Network Configuration Discovery	Enumerated network interfaces and listening services
T1046	Network Service Discovery	Reviewed open ports and active services using ss -tulnp
T1053.003	Scheduled Task/Job: Cron	Reviewed user and system cron jobs for persistence mechanisms
T1078	Valid Accounts	Reviewed local user accounts and privilege memberships
T1110	Brute Force	Examined authentication logs related to SSH access activity
T1021.004	Remote Services: SSH	Reviewed SSH daemon configuration and exposed SSH services

T1543	Create or Modify System Process	Reviewed enabled system services and startup configurations
T1069	Permission Groups Discovery	Examined sudo group membership and privilege assignments

Security Relevance

Mapping Linux administrative and audit activities to the MITRE ATT&CK framework helps security teams better understand how normal system configurations may be abused by attackers during different phases of an intrusion.

For example:

- Cron jobs may be abused for persistence
- Weak SSH configurations may allow unauthorized remote access
- Excessive sudo privileges may increase privilege escalation risk
- Open network services may expand attack surface exposure
- Poor logging visibility may reduce detection capability

Understanding these relationships improves defensive visibility and strengthens both security monitoring and hardening efforts.

Project Value

This mapping demonstrates practical blue team awareness by connecting Linux system administration reviews with real-world attacker techniques used during compromise attempts against Linux infrastructure.

The project reflects foundational SOC analyst, Linux security monitoring, and GRC-oriented assessment skills commonly required in cybersecurity operations and federal contractor environments.